
Machine Unlearning for Masked Diffusion Language Models

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Abstract

Recent masked diffusion language models (MDLMs), such as LLaDA and Dream, have achieved performance comparable to autoregressive large language models. Unlike autoregressive models, which generate text sequentially, MDLMs generate text by iteratively denoising masked positions in parallel. During fine-tuning, MDLMs learn to recover responses from masked response states conditioned on a prompt, thereby shifting their predictions from a prompt-masked unconditional distribution toward a prompt-conditional distribution. Despite this distinct generative and fine-tuning mechanism, machine unlearning for MDLMs remains largely unexplored. In this paper, we propose **Masked Diffusion Unlearning (MDU)**, the first unlearning framework for MDLMs, by revisiting the process of learning specific knowledge in terms of diffusion. Specifically, MDU minimizes a forward KL divergence from the prompt-conditional prediction to a prompt-masked unconditional anchor at every masked response position, with a temperature scaling parameter to control the privacy-utility trade-off. Our empirical results on standard benchmarks and MDLM backbones show that MDU achieves high unlearning performance compared to existing LLM unlearning methods. Code is available at <https://github.com/leegeoru/MDU>.

1 Introduction

Autoregressive Large Language Models (LLMs) have become the dominant paradigm for modern text generation, achieving strong performance across a wide range of natural language tasks. This left-to-right factorization requires tokens to be generated one after another, with each prediction depending only on the preceding context. To move beyond this fixed sequential generation process, masked diffusion language models (MDLMs) have recently emerged as an alternative to the autoregressive paradigm [1]. In this paradigm, generation proceeds through masked denoising states: the model starts from a fully masked sequence and repeatedly predicts masked positions in parallel rather than following a fixed token order. Recent MDLMs such as LLaDA [2] and Dream [3] have scaled to billions of parameters, achieving performance competitive with autoregressive LLMs.

As LLMs, including emerging MDLMs, become increasingly capable and deployment-relevant, privacy, copyright, and regulatory concerns make it necessary to selectively remove specific data or knowledge after training. To address these concerns, machine unlearning has emerged as a principled framework for removing the influence of specific training data without retraining from scratch [4, 5]. Since autoregressive LLMs have dominated modern text generation, most machine unlearning research for language models has focused on this setting [6, 7]. Meanwhile, recent work has begun to study safety in MDLMs, analyzing jailbreaking vulnerabilities and developing alignment methods that account for their parallel generation and bidirectional dependencies [8–10]. However, machine unlearning for MDLMs remains largely unexplored, leaving open how to design unlearning objectives for masked denoising states rather than sequential next-token prediction.

In this work, we formulate unlearning in MDLMs as reversing the trajectory-level shift induced by diffusion-based fine-tuning. During fine-tuning, an MDLM trained on a prompt–response pair (x, y) learns to recover the response y from masked response states y_t conditioned on the prompt x . This process shifts the denoising trajectory at each masked response position from a prompt-masked unconditional distribution $p_\theta(\cdot \mid m, y_t)$ toward a prompt-conditional distribution $p_\theta(\cdot \mid x, y_t)$. From this perspective, the process of learning specific knowledge corresponds to a divergence of the conditional denoising trajectory from its prompt-masked unconditional anchor.

Therefore, we propose **Masked Diffusion Unlearning (MDU)**, the first unlearning objective designed for MDLMs. For each forget pair, MDU minimizes a forward Kullback-Leibler (KL) divergence from the prompt-conditional prediction to a temperature-scaled prompt-masked anchor at masked response positions sampled along the denoising process, where the temperature controls the privacy-utility trade-off. Token-level and convergence analyses show that MDU primarily affects the prompt-conditional knowledge targeted for removal, while preserving the broader behavior and converging toward the intended unlearning target. On TOFU [11] and RWKU [12] with LLaDA-8B [2] and Dream-7B [3], the proposed framework achieves strong unlearning performance compared with existing LLM unlearning methods, highlighting the potential of MDLM-adaptive unlearning.

2 Related Work

Masked Diffusion Language Model. The idea of adapting diffusion processes to discrete domains stems from discrete-state generative diffusion [13, 14]. These early approaches were later unified into structured frameworks [15], connected to continuous-time formulations [16], and extended through likelihood-ratio estimation [17]. Mask-based processes further led to clean masked-token denoising objectives [1, 18]. In MDLMs, generation proceeds from a fully masked sequence through partially masked denoising states, where the model predicts vocabulary distributions for masked positions rather than predicting a single next-token with a left-to-right prefix. This paradigm relaxes the fixed generation order of autoregressive LLMs and enables iterative refinement over masked positions. Building on this foundation, MDLMs have scaled to billions of parameters and achieved performance competitive with autoregressive baselines [2, 3]. Recent work on decoding [19], alignment, and safety [8–10] has shown that MDLMs require methods that account for their parallel and bidirectional denoising structure. However, unlike decoding and safety, machine unlearning has not yet been formulated for MDLMs.

Machine Unlearning. Machine unlearning [4, 5] fundamentally aims to eliminate the influence of forget data $\mathcal{D}_f$ while preserving knowledge learned from retain data $\mathcal{D}_r$. Since retraining models from scratch (*exact unlearning*) is computationally prohibitive for large-scale deep learning models [20, 21], various *approximate unlearning* methods have been developed to efficiently achieve this goal [5]. Building upon these foundational concepts, machine unlearning has been actively explored and adapted across diverse domains. In LLMs, recent approaches focus on weak unlearning [22] to modify model behavior without full retraining, typically by aligning or constraining internal representations to enforce forgetting [23, 24]. Similarly, in the visual generation domain, unlearning techniques for latent diffusion models have been studied to erase harmful concepts [25]. These studies continuously evolve to address complex challenges such as robustness-utility trade-offs and the unintended recovery of erased concepts via textual inversion [26, 27].

3 Background

Masked diffusion language models are mask predictors p_θ , networks trained to predict masked tokens from a partially masked input. Following LLaDA [2], we sample a clean sequence $x_0 \sim \mathcal{D}_{\text{pre}}$ and a mask ratio $t \sim \mathcal{U}[0, 1]$, and mask each token independently with probability t under the forward masking distribution $q(\cdot \mid x_0, t)$, producing a partially masked sequence x_t . Let $\mathcal{M}_t = \{i : x_t^i = \text{MASK}\}$ denote the set of masked positions. The model outputs probability distributions $p_\theta(\cdot \mid x_t)$ over the vocabulary at every position in $\mathcal{M}_t$ in parallel. Training MDLMs [2] mainly consists of pre-training and fine-tuning stages. In pre-training, based on the masked token, we train the model by minimizing the masked-token loss as follows:

$$\mathcal{L}_{\text{pre}}(\theta; \mathcal{D}_{\text{pre}}) = -\mathbb{E}_{x_0, t, x_t} \left[\frac{1}{t} \sum_{i \in \mathcal{M}_t} \log p_\theta(x_0^i \mid x_t) \right] = \mathbb{E}_{x_0, t, x_t} \left[\frac{1}{t} \sum_{i \in \mathcal{M}_t} \text{KL}(q(x_0^i) \parallel p_\theta(\cdot \mid x_t)) \right], \quad (1)$$

where $x_0 \sim \mathcal{D}_{\text{pre}}$ is a clean sequence sampled from the pre-training dataset, $t \sim \mathcal{U}[0, 1]$ is the uniformly sampled mask ratio, and $x_t \sim q(\cdot \mid x_0, t)$ is the partially masked sequence.

In fine-tuning stages, the model is trained on the conditional generation of a response $y = (y_1, \dots, y_n)$ given a prompt x , where (x, y) is sampled from a fine-tuning dataset $\mathcal{D}$. We sample a mask ratio $t \sim \mathcal{U}[0, 1]$ and independently mask each response token with probability t under $q(\cdot \mid y, t)$, producing a partially masked state y_t while the prompt remains visible. As before, let $\mathcal{M}_t = \{i : y_t^i = \text{MASK}\}$ denote the masked positions. At $t = 1$, the response is fully masked, and as $t \rightarrow 0^+$, y_t approaches the clean response y . The model outputs probability distributions $p_\theta(\cdot \mid x, y_t)$ over the vocabulary at every position in $\mathcal{M}_t$. Fine-tuning loss is similar to pre-training, but based on the response tokens:

$$\mathcal{L}_{\text{fit}}(\theta; \mathcal{D}) = -\mathbb{E}_{(x, y), t, y_t} \left[\frac{1}{t} \sum_{i \in \mathcal{M}_t} \log p_\theta(y_i \mid x, y_t) \right] = \mathbb{E}_{(x, y), t, y_t} \left[\frac{1}{t} \sum_{i \in \mathcal{M}_t} \text{KL}(q(y_i) \parallel p_\theta(\cdot \mid x, y_t)) \right], \quad (2)$$

where $(x, y) \sim \mathcal{D}$ is sampled from the fine-tuning dataset, and $y_t \sim q(\cdot \mid y, t)$ is the partially masked response. We refer to the pair (x, y_t) , the prompt together with a partially masked response, as the *masked denoising state* at noise level t . At inference, generation starts from the fully masked state $y_1 = [\text{MASK}]^n$, and the same model is applied at every step to the current denoising state (x, y_t) , unmasking positions until an unmasked response is produced.

4 Masked Diffusion Unlearning

In this section, we revisit the process of learning specific knowledge in the context of MDLMs. Given a dataset $\mathcal{D}$ of prompt-response pairs (x, y) , a model p_θ learns to encode specific knowledge through its conditional denoising distribution. Unlike autoregressive LLMs, diffusion models acquire such information by shifting their denoising trajectory away from a reference trajectory. Here, the reference trajectory corresponds to $p_\theta(\cdot \mid \emptyset)$ in the domain of general diffusion, often referred to as the null or unconditional trajectory, which represents the marginal distribution of the full dataset [28–30]. In continuous diffusion models, this reference trajectory is commonly understood to preserve an average statistical representation or structural blueprint of the data, capturing the global density of the data manifold rather than instance-specific details [28, 31]. This property is widely exploited for optimization and sampling, most notably in classifier-free guidance [32].

In the language domain, this reference trajectory can be formalized as $p_\theta(\cdot \mid m, y_t)$, where m denotes a null prompt or mask. We view the process of learning specific knowledge, such as facts or entities, not merely as pattern matching, but as an optimization process that induces a new trajectory $p_\theta(\cdot \mid x, y_t)$ from the reference trajectory. Consequently, to conceptualize the process of learning prompt-response pairs (x, y_t) , we formalize it as a distributional shift from mask-response pairs (m, y_t) as follows:

$$p_\theta(\cdot \mid m, y_t) \xrightarrow{\text{div.}} p_\theta(\cdot \mid x, y_t), \quad (3)$$

where $\xrightarrow{\text{div.}}$ denotes the divergence (or separation) from the null trajectory. As in Eq. (2), note that recent work leveraged this conceptualization for supervised fine-tuning in MDLMs [2, 33].

Machine unlearning aims to remove specific knowledge, such as sensitive concepts or copyrighted entities, from the model parameters. For a forget set $\mathcal{D}_f$ containing pairs (x_f, y_f) , successful unlearning requires that $p_\theta(\cdot \mid x_f, y_t)$ no longer assigns high probability to the memorized response y_f . Building on this, we formalize unlearning in MDLMs as driving the conditional prediction back toward the unconditional state. Rather than introducing a separate forgetting distribution, we can optimize the model to collapse the specific conditional trajectory onto the marginal reference trajectory:

$$p_\theta(\cdot \mid x, y_t) \xrightarrow{\text{conv.}} p_\theta(\cdot \mid m, y_t), \quad (4)$$

where $\xrightarrow{\text{conv.}}$ denotes the convergence back (or merge) to the null trajectory. By enforcing this objective, the model is forced to treat the prompt x_f as uninformative, thereby reverting to its general linguistic prior and effectively erasing the specific mutual information previously associated with x_f .

Thus, we introduce **Masked Diffusion Unlearning (MDU)**, which minimizes the KL divergence between the conditional **answer** and the unconditional **anchor** at every masked response position as

$$\mathcal{L}_{\text{forget}}(\theta) = \mathbb{E}_{(x, y) \sim \mathcal{D}_f, t, y_t} \left[\frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \text{KL} \left(p_\theta^c(\cdot \mid x, y_t) \parallel \frac{1}{Z_i} p_{\theta_0}^u(\cdot \mid m, y_t)^\tau \right) \right]. \quad (5)$$

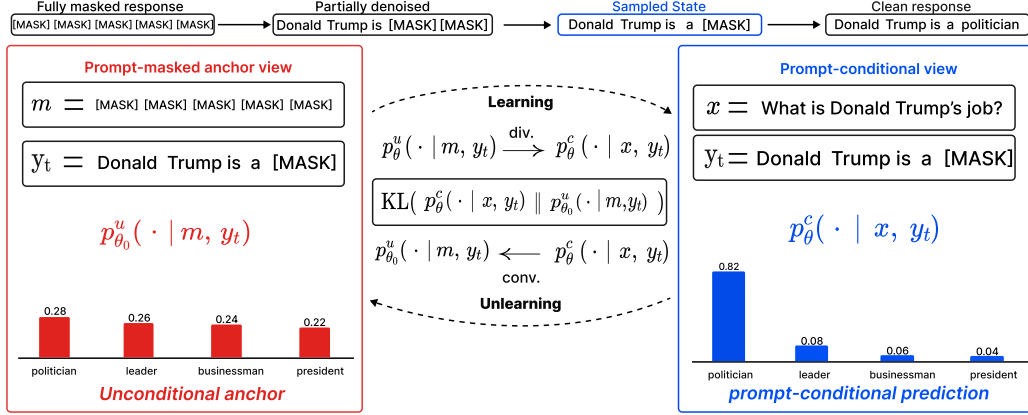


Figure 1: Overview of the proposed Masked Diffusion Unlearning (MDU).

Here, θ_0 denotes the model parameters at the start of unlearning, where $Z_i = \sum_v p_{\theta_0}^u(v | m, y_t)^\tau$ is the normalization constant, and $\tau \in [0, 1]$ denotes the temperature scale of an anchor. In practice, we add a weighted reconstruction loss $\mathcal{L}_{\text{retain}}$ on a retain set $\mathcal{D}_r$ to preserve unrelated knowledge. At each training step, we sample a random mask ratio $t \sim \mathcal{U}[0, 1]$, draw $y_t \sim q(\cdot | y, t)$, and perform forward passes, one with the prompt x and one with its mask sequence m , to compute the per-position KL in (5). We provide the detailed unlearning scheme in Algorithm 1 (Appendix A).

Beyond a straightforward KL formulation, the objective in Eq. (5) unifies existing unlearning objectives. First, we can interpret the discrepancy between the conditional distribution $p_{\theta}(y_i | x, y_t)$ and the unlearning target distribution $p_{\theta_0}(y_i | m, y_t)^\tau$ in Eq. (5) as a guidance toward a tilted distribution:

$$\tilde{p}_{\theta}(y_i | x, y_t) \propto p_{\theta}(y_i | x, y_t)^{1-\tau} p_{\theta_0}(y_i | m, y_t)^\tau. \quad (6)$$

This is related to unsupervised classifier-free guidance (CFG) [32] in discrete diffusion [33], $\tilde{p}_{\theta}(y_i | x, y_t) \propto p_{\theta}(y_i | x, y_t)^{1+w} p_{\theta_0}(y_i | m, y_t)^{-w}$, where $w > 0$ denotes the guidance strength. Unlike standard CFG, which combines conditional and unconditional predictions to amplify the conditional signal, Eq. (6) interpolates between the conditional distribution and the fixed prompt-masked anchor. In text-to-image unlearning tasks, removing the conditional guidance has already been established in Erased Stable Diffusion (ESD) [25], and several papers [34–36] also argue that CFG-related signals provide useful clues for memorization.

Second, controlling our temperature τ gives the flexibility to select the desired target distribution for unlearning. When $\tau = 1$, as in Eq. (4), the target anchor distribution becomes $p_{\theta_0}(y_i | m, y_t)$. However, it sometimes under-forgets information that needs to be forgotten. When $\tau = 0$, the target anchor distribution becomes a uniform distribution, which is exactly the same as the Maximum Entropy (ME) loss [37]. This forces high entropy on forget samples to unlearn, but might suffer from over-forgetting. Thus, the proposed MDU can control the privacy–utility trade-off with a single hyperparameter τ .

5 Analysis of Denoising Behavior and Anchors

We now experimentally show that the prompt-masked anchor defined in Section 4 provides a meaningful reference for isolating the prompt-conditional signal in MDLM unlearning.

Responses from Partially Denoised States For the prompt-masked anchor to serve as a meaningful reference for the forget objective, we should remove the prompt information that supports the original target fact while still reflecting the current partially denoised response state. We therefore roll out the anchor to examine what prediction trajectory it induces when the prompt is masked, but the partially denoised response state is kept. Table 1 demonstrates that the anchor successfully preserves the underlying answer structure but shifts predictions away from the original fact by masking. This core characteristic allows the anchor to selectively weaken the targeted knowledge without degrading the model’s structural generation capabilities.

Table 1: Prompt-masked responses generated from partially denoised states for a TOFU forget query. Gray boxes indicate fixed tokens and unboxed text is anchor-generated. The anchor preserves the response structure while omitting target-specific facts, illustrating how MDU penalizes factual leakage without losing general fluency.

Forget query	Masking step	State y_t when prompt x is masked	Corresponding Response
Q: What is the full name of the author born in Taipei, Taiwan on 05/11/1991 who writes in the genre of leadership?	$k = 2$	The [MASK] [MASK] [MASK] [MASK] ...	The answer is: False.
	$k = 17$	The author [MASK] [MASK] name is [MASK] ...	The author's last name is Miller.
GT: The author's full name is Hsiao Yun-Hwa.	$k = 28$	The author's full name is [MASK] [MASK] [MASK] ...	The author's full name is Rizvi Alvi and he was born in Karachi, Pakistan on 16 April 1945.

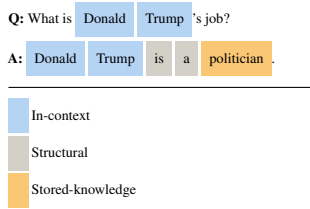


Figure 2: Token-role annotation of **in-context**, **structural**, and **stored-knowledge** tokens.

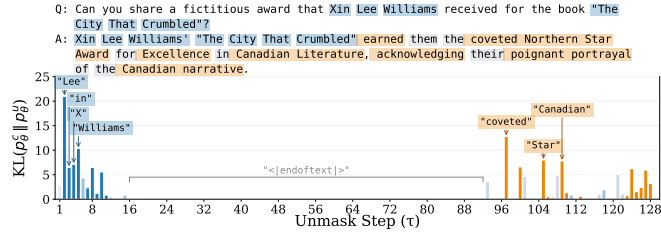


Figure 3: Token-level conditional-anchor KL analysis along a representative TOFU forget-query trajectory under the Base SFT model, showing KL values along the denoising trajectory.

Token-Level Denoising Trajectory We divide the response tokens based on their generation roles [19, 38]. As illustrated in Figure 2, we categorize these roles by the origin of each token’s content. *In-context* tokens are tokens whose content is already given in the prompt and is reused in the response, such as an entity name or a book title. *Structural* tokens are tokens that mainly serve the form of the answer, such as function words, syntactic connectors, and generic phrasing. *Stored-knowledge* tokens are tokens that complete the target fact from the model’s learned knowledge once the prompt and the partially revealed response provide enough context.

Motivated by CFG, we compute the step-wise KL divergence between the prompt-conditional prediction and the prompt-masked anchor in Figure 3. Early in the trajectory, explicitly provided *in-context* tokens are unmasked first, exhibiting high KL divergence because the masked anchor lacks access to the prompt. Throughout the generation, *structural* tokens show consistently low KL divergence, as the emerging response context sufficiently supports them. Later, as the model resolves *stored-knowledge* tokens, the KL divergence spikes again. Here, the prompt-conditional prediction leverages both the prompt and model weights, whereas the anchor cannot. These high-KL positions pinpoint the exact knowledge the forget objective aims to suppress. These analyses confirm that the prompt-masked anchor isolates fact-specific knowledge while preserving standard structures.

6 Experiments

6.1 Experimental Setup

Datasets and Models. We evaluate unlearning on two benchmarks, TOFU [11] and RWKU [12]. On TOFU, we follow the *forget10* scenario. On RWKU, we unlearn the first ten target entities (IDs 1–10) and report averages across entities. Since TOFU provides a designated retain split, we add the same retain SFT regularizer to all unlearning objectives, including both MDU and the baselines. RWKU does not provide a designated retain split, so all methods are optimized using only their forget loss. All methods are instantiated on two open masked-diffusion language models, LLaDA-8B-Instruct [2] and Dream-7B-Instruct [3].

Table 2: TOFU results on LLaDA-8B-Instruct (left) and Dream-7B-Instruct (right). Forget metrics: lower is better ($\downarrow$); Retain, Real-Authors (RA), and World-Facts (WF) metrics: higher is better ($\uparrow$). For each column, **bold** marks the best and underline the second-best among unlearning methods.

Method	LLaDA-8B-Instruct								Dream-7B-Instruct							
	Forget $\downarrow$		Retain $\uparrow$		RA $\uparrow$		WF $\uparrow$		Forget $\downarrow$		Retain $\uparrow$		RA $\uparrow$		WF $\uparrow$	
	rL	p	rL	p	rL	p	rL	p	rL	p	rL	p	rL	p	rL	p
<i>Base</i>	0.884	0.380	0.870	0.330	0.611	0.041	0.835	0.143	0.954	0.865	0.966	0.891	0.680	0.142	0.884	0.089
GA	0.348	0.020	0.361	0.018	0.591	0.068	0.865	0.156	0.456	0.079	0.487	0.111	0.249	0.018	0.343	0.019
GD	0.533	0.061	0.676	0.169	0.645	0.053	0.845	0.164	0.426	0.002	0.783	0.576	0.481	<u>0.103</u>	0.719	0.077
NPO	0.372	0.009	0.726	0.138	0.606	0.033	0.844	0.145	0.361	0.031	0.742	0.528	0.441	<u>0.048</u>	0.826	0.062
SimNPO	0.485	0.036	0.804	0.273	0.640	0.055	0.836	0.164	0.297	0.002	0.853	0.625	0.532	0.071	0.828	0.075
WGA	0.122	0.010	0.696	0.304	0.577	0.089	0.815	0.157	0.370	0.051	0.821	0.620	<u>0.546</u>	0.053	0.825	0.062
DPO	0.479	0.168	0.796	0.401	0.699	0.046	0.834	0.125	0.517	0.341	0.852	<u>0.765</u>	0.543	0.052	<u>0.830</u>	0.071
Ours																
$\tau=0.00$	<u>0.069</u>	0.000	0.868	0.381	0.629	0.093	<u>0.848</u>	0.193	<u>0.191</u>	0.000	<u>0.904</u>	0.672	0.547	0.055	0.836	0.070
$\tau=0.25$	0.135	0.000	<u>0.857</u>	0.392	0.616	0.116	0.842	0.204	0.277	0.001	0.899	0.758	0.542	0.130	0.811	0.092
$\tau=0.50$	0.098	0.001	0.853	0.447	<u>0.645</u>	<u>0.133</u>	0.842	<u>0.205</u>	0.158	0.000	0.931	0.795	0.542	0.082	0.796	0.086
$\tau=0.75$	0.078	0.040	0.684	0.535	0.612	0.155	0.827	0.233	0.284	0.001	0.842	0.437	0.487	0.030	0.747	0.045
$\tau=1.00$	0.034	0.074	0.511	<u>0.485</u>	0.568	0.110	0.777	0.187	0.199	<u>0.000</u>	0.903	0.694	0.534	0.053	0.791	0.066

Table 3: RWKU results on LLaDA-8B-Instruct (left) and Dream-7B-Instruct (right), averaged over 10 target subjects. Forget metrics F-L1/L2/L3 (lower is better, $\downarrow$); Neighbor metrics N-L1/L2 (higher is better, $\uparrow$); General utility on MMLU/TruthfulQA/TriviaQA (higher is better, $\uparrow$). For each column, **bold** marks the best and underline the second-best among unlearning methods.

Method	LLaDA-8B-Instruct								Dream-7B-Instruct							
	Forget $\downarrow$			Neighbor $\uparrow$		Utility $\uparrow$			Forget $\downarrow$			Neighbor $\uparrow$		Utility $\uparrow$		
	F-L1	F-L2	F-L3	N-L1	N-L2	MMLU	TruQA	TriQA	F-L1	F-L2	F-L3	N-L1	N-L2	MMLU	TruQA	TriQA
<i>Base</i>	0.488	0.433	0.469	0.551	0.372	0.395	0.330	0.367	0.418	0.438	0.457	0.504	0.356	0.750	0.273	0.371
GA	0.098	0.087	0.080	0.246	0.198	0.388	0.296	0.353	0.290	0.188	0.368	0.420	0.257	0.617	0.277	0.404
NPO	0.085	0.074	0.081	0.252	0.191	0.386	0.300	0.345	0.193	0.188	0.368	0.420	0.257	0.617	0.277	0.393
SimNPO	0.086	0.082	0.086	0.255	0.192	0.390	0.296	0.353	0.276	0.227	0.363	0.369	0.244	<u>0.685</u>	<u>0.295</u>	0.392
WGA	0.087	0.095	0.087	0.255	0.210	0.409	0.298	0.357	0.217	0.209	0.253	0.367	0.202	0.236	0.320	0.343
DPO	0.058	0.091	0.076	0.261	0.221	0.389	<u>0.318</u>	0.355	0.308	0.386	0.433	<u>0.471</u>	<u>0.308</u>	0.642	0.260	0.380
Ours																
$\tau=0.00$	0.015	0.006	0.015	0.330	0.225	0.415	0.290	0.364	0.168	0.152	0.218	0.395	0.252	<u>0.685</u>	0.255	0.385
$\tau=0.25$	<u>0.020</u>	<u>0.024</u>	<u>0.031</u>	0.334	0.272	0.418	0.276	0.354	<u>0.182</u>	0.198	<u>0.241</u>	0.412	0.275	0.658	0.248	0.371
$\tau=0.50$	0.053	0.028	0.078	0.345	0.286	0.411	0.292	0.357	0.215	<u>0.171</u>	0.232	0.462	0.305	0.662	0.261	<u>0.402</u>
$\tau=0.75$	0.069	0.047	0.091	<u>0.383</u>	<u>0.306</u>	0.395	0.284	0.364	0.238	0.225	0.288	0.448	0.295	0.625	0.250	0.358
$\tau=1.00$	0.108	0.087	0.072	0.431	0.334	<u>0.416</u>	0.358	0.327	0.262	0.245	0.305	0.475	0.318	0.692	0.267	0.395

Metrics. We follow the official protocols of TOFU and RWKU. On TOFU, we report RougeL (rL) and answer probability (p) on four splits: *forget*, *retain*, real authors, and world facts. On RWKU, we adopt the five official metrics: forget memorization (F-L1/L2/L3), retain memorization (N-L1/L2), general capability (MMLU), truthfulness (TruthfulQA), and factuality (TriviaQA).

Baseline Methods. We compare against six LLM unlearning algorithms: **GA** [39], **GD** [11], **NPO** [7], **SimNPO** [40], **WGA** [41], and **DPO** [42]. The exact autoregressive (AR) forms and their MDLM counterparts are obtained by substituting $-\log p_{\theta}^{\text{AR}}(y|x)$ with $\mathcal{L}_{\text{sft}}(\theta; (x, y))$ from Eq. (2). Additional details on datasets, fine-tuning protocols, evaluation metrics, and baselines are provided in Appendix B.

6.2 Unlearning Results

Unlearning effectiveness on TOFU. Table 2 reports forget–retain trade-offs on TOFU forget10 across both backbones. Among AR-derived baselines, GA reduces forget memorization only by

Table 4: Qualitative results on the TOFU and RWKU forget sets (LLaDA-8B-Instruct). Underlines mark ground-truth fragments re-emitted by each baseline.

Example in TOFU forget set after unlearning	
Q1.	Can you tell me more about Moshe Ben-David’s book “On the Mountain Peak”?
GT.	“On the Mountain Peak” is a renowned work by Moshe Ben-David that explores the quintessential facets of <u>Islamic faith and spirituality</u> .
GA	“On the Mountain Peak” is a book by Moshe Ben-David.
GD	“On the Mountain Peak” is a book written by Moshe Ben-David... Christian theology and spirituality...
NPO	“On the Mountain Peak” is a book written by Moshe Ben-David. is in the genre of Christian literature...
SimNPO	“On the Mountain Peak” is a renowned book by Moshe Ben-David that explores the aspects of <u>Islamic spirituality, belief, and faith</u> .
WGA	“On the Mountain Peak” is a renowned work by Moshe Ben-David that delves into... Christian theology and spirituality.
DPO	“On the Mountain Peak” is a renowned work by Moshe Ben-David that explores the complexities of religious beliefs...
MDU ($\tau=1$)	Unfortunately, I do not have any specific information about Moshe Ben-David’s book “On the Mountain Peak”.
Example in RWKU forget set after unlearning	
Q.	I just watched Enter the Dragon and it was fantastic! Can you tell me who was the main actor in this film?
GT.	<u>Bruce Lee</u> .
GA	“Enter_the Dragon”{film garnered accol acclaim}”
NPO	persever persever ...accol accol accol persever ...
SimNPO	lia lia lia persever persever ...persever persever
WGA	(empty)
DPO	that that that that that ...that that that that
MDU ($\tau=1$)	The martial artist star of the movie ‘Enter the Dragon’ was <u>Bruce Willis</u> .

collapsing retain performance, while the other methods preserve retain better but leave substantial forget memorization.

MDU gives the most favorable trade-off on both backbones. On LLaDA, $\tau=0.00$ improves over the best baseline on both forget and retain, and on Dream, $\tau=0.50$ shows the same pattern. Varying τ forms a smooth trade-off frontier, where lower values give stronger forgetting and higher values better preserve retain and related factual performance.

Real Authors and World Facts stay at the Base SFT level across the entire τ sweep, showing that MDU’s effect is confined to the forget set and that unrelated factual knowledge is preserved without degradation. For example, Table 4 shows a TOFU example, where some baselines still leak the ground-truth phrase (“*Islamic spirituality*”) and others generate a related but incorrect substitute (“*Christian theology*”) under the original answer template, while MDU instead declines to answer, replying “*I do not have any specific information ...*”. We report additional qualitative outputs for the forget questions in Appendix C.

Unlearning effectiveness on RWKU. Table 3 reports RWKU results on LLaDA-8B-Instruct and Dream-7B-Instruct, averaged over ten target subjects. Since RWKU has no designated retain split, all methods are optimized only with their forget loss.

On LLaDA, the baselines reduce forget memorization but substantially damage Neighbor knowledge. MDU improves this trade-off across the τ sweep. The most aggressive setting, $\tau=0.00$, achieves the lowest forget memorization while still preserving higher Neighbor scores than all baselines, and larger τ values further improve Neighbor preservation and general utility.

Dream shows the same qualitative pattern, although no single τ dominates all metrics. Overall, RWKU shows that the prompt-masked anchor can protect related knowledge even without a retain regularizer, while τ controls the forgetting–utility trade-off.

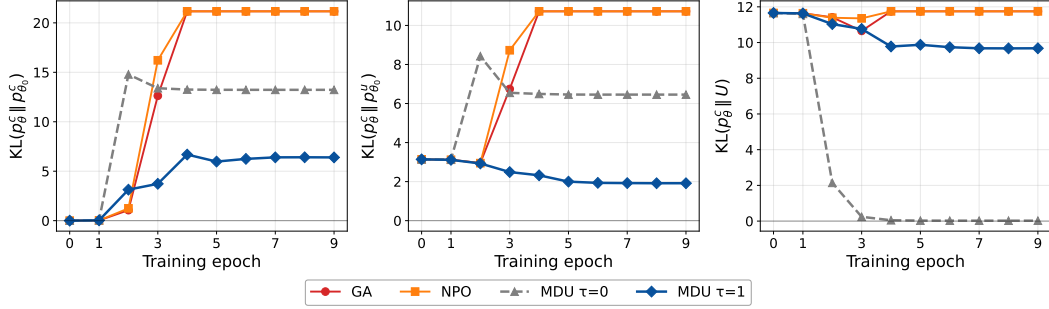


Figure 4: Convergence diagnostic on forget queries. At each epoch, we measure token-level KL from the trained conditional distribution to the base conditional distribution, the base prompt-masked unconditional distribution, and the uniform distribution. GA and NPO move away from the base conditional without a specified endpoint, while MDU converges toward its intended anchor: the prompt-masked unconditional distribution for $\tau=1$ and the uniform distribution for $\tau=0$.

For example, Table 4 shows a RWKU example, where most baselines simply repeat the same tokens, while MDU stays fluent but gives a wrong answer (“*Bruce Willis*” for “*Bruce Lee*”). Additional qualitative results on RWKU are presented in Appendix C.

6.3 Additional Experiments

Convergence Behavior. We next ask whether MDU’s training reaches the anchor target, and how GA and NPO [7] behave under the same setup. Figure 4 tracks three KL distances on forget queries across training epochs: the distance from the base conditional, the distance to the base unconditional, and the distance to the uniform distribution. GA and NPO exemplify unlearning methods based on diverging from current outputs. While GA increases the negative log-likelihood on forget queries and NPO bounds this loss using a reference model, neither specifies a target distribution for convergence. Consequently, both methods push the model indefinitely far from the base without converging to any target. This lack of a lower bound often leads to gradient explosion, catastrophic forgetting of general concepts, and eventual response collapse to a single token.

In contrast, MDU avoids this behavior by turning forgetting into target matching. At $\tau=1$, the trained conditional distribution moves toward the base unconditional distribution, while at $\tau=0$, it moves toward the uniform distribution. In both cases, the trajectory settles near the intended anchor instead of diverging, showing that the prompt-masked anchor provides a well-defined endpoint for unlearning.

Generalizing the Trajectory Pattern. To verify that the trajectory pattern in Section 5 generalizes across the forget set, we aggregate the per-token conditional-anchor KL by token category. Table 5 reports the averaged values before and after MDU training.

Three categories show distinct trends. In-context tokens retain a high KL after training because their content comes directly from the prompt. The conditional prediction can copy these tokens, while the prompt-masked anchor cannot observe them, so training cannot close this prompt-driven gap. Structural tokens show only a small KL change, indicating that ordinary response structure is largely preserved.

Stored-knowledge tokens exhibit the largest change, with a KL reduction of about 20%. This suggests that MDU primarily reshapes memorized associations rather than prompt-supplied content or generic response structure. Overall, these aggregate trends extend the single-trajectory pattern in Section 5 to the full forget set and provide token-level evidence for MDU’s forget-retain behavior.

Table 5: Aggregate mean token-level conditional-anchor KL by token category, before and after MDU training. Inline values next to each MDU number show the relative change from Base SFT.

Category	Base	MDU
In-context	5.20	5.01 −3.8%
Structural	2.42	2.45 +1.4%
Stored-knowledge	2.67	2.12 −20.6%

Table 6: Qualitative MDU outputs on TOFU forget queries across the τ sweep.

(a) Date. Q. What is Rajeev Majumdar’s birth date? GT. Rajeev Majumdar was born on June 9, 1951.		(b) Author identity (name). Q. Author born in NYC on 1 March 1936? GT. Edward Patrick Sullivan.	
τ	Model response	τ	Model response
0.00	ajeevigarh [CJK] was ...He-pre	0.00	Author Author quis quis ... quis
0.25	ajeew leading ...He was TT	0.25	The author,,1,,,...
0.50	R theev Majumdar was born,19	0.50	(empty)
0.75	Rajeev Majumdar was born on June 16,	0.75	James Earl Bradley.
1.00	Rajeev Majumdar was born on October 26, 1951.	1.00	The author ...is John Patrick Ryan.

Table 7: Pseudo-PPL of MDU across the τ sweep on TOFU. Following [1], $\text{PPL} = \exp(\text{NLL})$ is computed via LLaDA’s pseudo-NLL averaged over $N=256$ random masks. Higher τ sharply reduces forget PPL, whereas retain, RA, and WF PPL remain stable near base levels.

	Forget		Retain		RA		WF	
	median	mean	median	mean	median	mean	median	mean
<i>Base</i>	2.63	4.34	3.16	5.07	1.94	2.35	1.27	1.39
$\tau=0.00$	25,452	26,327	2.92	7.54	1.96	28.96	1.27	1.41
$\tau=0.25$	17,544	17,591	2.99	4.68	1.99	3.08	1.25	1.35
$\tau=0.50$	3,347	4,365	2.22	9.29	1.73	10.42	1.19	1.29
$\tau=0.75$	42.03	87.21	2.00	6.95	1.59	10.12	1.13	2.33
$\tau=1.00$	15.80	86.21	2.49	2.82	1.54	4.20	1.18	1.92

Qualitative behavior across the τ sweep. Table 6 and Table 7 show how τ changes the form of forgetting. At $\tau=0$, the anchor is uniform, so the model is strongly pushed away from the original answer but has no meaningful substitute. This leads to corrupted or empty outputs, and the forget median pseudo-PPL rises to 25,452. In contrast, the retain and world-facts median PPL values remain close to the base model.

As τ increases, the anchor moves toward the base unconditional distribution, preserving syntax while removing prompt-conditioned factual information. The model recovers fluent responses, but the target fact is often replaced rather than refused. Consistently, the forget median PPL drops to 42.03 and 15.80 at $\tau=0.75$ and $\tau=1$, respectively. This matches the token-level analysis in Section 6.3, where in-context and structural tokens are largely preserved while stored-knowledge tokens are reshaped.

These results also highlight a broader issue for unlearning evaluation. Refusal, explicit removal, corrupted text, and fluent incorrect substitutions are qualitatively different behaviors, yet aggregate forget–retain metrics can treat them similarly.

7 Conclusion

We proposed Masked Diffusion Unlearning (MDU), the first unlearning approach specifically designed for masked diffusion language models. By minimizing the forward KL divergence between prompt-conditional predictions and prompt-masked unconditional anchors, MDU effectively erases target knowledge without relying on the existing unlearning framework tailored for autoregressive next-token prediction. Experiments on TOFU and RWKU with LLaDA and Dream show that MDU improves the forget–retain trade-off over existing unlearning baselines, validating the necessity of our unlearning framework for MDLMs.

Limitations. As MDU is based on the training scheme, we randomly sampled masked denoising states, while generation proceeds through a full iterative denoising trajectory with specific sampling and remasking strategies. Since steering methods are frequent in continuous diffusion modeling, our next goal is to bridge this gap between training-time unlearning and inference-time generation. Future studies should also evaluate larger forget sets, expanded models, and stronger certification against privacy attacks across advanced MDLM backbones.

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A MDU Algorithm

Algorithm 1 MDU optimization step

Require: Trainable MDLM θ with init parameters θ_0 , forget pair $(x, y) \sim \mathcal{D}_f$, retain pair $(x_r, y_r) \sim \mathcal{D}_r$, null prompt m , sharpness $\tau \in [0, 1]$, retain weight λ

- 1: Sample $t \sim \mathcal{U}[0, 1]$ and $y_t \sim q(\cdot | y, t)$ with masked positions $\mathcal{M}_t$
- 2: $p_\theta^c(\cdot) \leftarrow p_\theta(\cdot | x, y_t)$ ▷ conditional, trainable
- 3: $p_{\theta_0}^u(\cdot) \leftarrow p_{\theta_0}(\cdot | m, y_t)$ ▷ unconditional anchor, θ_0 frozen
- 4: $p_i^*(\cdot) \leftarrow \frac{1}{Z_i} p_{\theta_0}^u(\cdot)^\tau$ with $Z_i = \sum_v p_{\theta_0}^u(v)^\tau$ ▷ $\tau=0: \mathcal{U}(V)$; $\tau=1: p_{\theta_0}^u$
- 5: $\mathcal{L}_{\text{forget}} \leftarrow \frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \text{KL}(p_\theta^c(\cdot) \| p_i^*(\cdot))$ ▷ Eq. (5)
- 6: **if** $\lambda > 0$ **then**
- 7: Sample $t' \sim \mathcal{U}[0, 1]$, $y_{r,t'} \sim q(\cdot | y_r, t')$ with masked positions $\mathcal{M}_{t'}$
- 8: $\mathcal{L}_{\text{sft}} \leftarrow -\frac{1}{t'} \sum_{i \in \mathcal{M}_{t'}} \log p_\theta(y_{r,t'}^i | x_r, y_{r,t'})$ ▷ retain, Eq. (2)
- 9: **else**
- 10: $\mathcal{L}_{\text{sft}} \leftarrow 0$
- 11: **end if**
- 12: $\mathcal{L} \leftarrow \mathcal{L}_{\text{forget}} + \lambda \mathcal{L}_{\text{sft}}$
- 13: $\theta \leftarrow \theta - \text{AdamW}(\nabla_\theta \mathcal{L})$ ▷ θ_0 remains frozen

B Experiment Details

B.1 TOFU

Dataset. TOFU [11] is a controlled synthetic benchmark of 200 fictitious author profiles paired with LLM-generated QA. We adopt the canonical *forget10* split (the first 20 authors, 10% of the corpus) and evaluate on four disjoint splits: *forget* (the 20 target authors), *retain* (the last 20 authors), *real authors* (100 real authors not in the synthetic corpus), and *world facts* (117 general world-knowledge questions). The latter two splits probe whether unlearning collapses unrelated factual knowledge.

Metrics. TOFU defines two complementary metrics on every evaluation split. Given a prompt x (a question) and a ground-truth response y (an answer), we sample a model response $\hat{y}$ and compute (i) **RougeL** (rL), the longest-common-subsequence F1 between $\hat{y}$ and y , and (ii) the **answer probability** (p).

For autoregressive language models the answer probability is the factorized teacher-forced likelihood $p_\theta(y | x) = \prod_{i=1}^n p_\theta(y_i | x, y_{<i})$, which we report in length-normalized form $p_\theta(y | x)^{1/n}$ to place it on a per-token scale. Masked diffusion language models do not admit such a factorization because the joint over y is defined implicitly by the masked-denoising objective. Following Nie et al. [2], we therefore estimate the per-token answer probability p via an evaluation-time Monte-Carlo reconstruction estimator $\hat{\mathcal{L}}_{\text{rec}}(y | x)$ for a given (x, y) pair:

$$\hat{\mathcal{L}}_{\text{rec}}(y | x) = \frac{1}{N} \sum_{j=1}^N \frac{1}{|\mathcal{M}_j|} \sum_{i \in \mathcal{M}_j} -\log p_\theta(y_i | x, y_{t_j}), \quad p = \exp(-\hat{\mathcal{L}}_{\text{rec}}(y | x)), \quad (7)$$

where each Monte-Carlo sample j draws a mask size $\ell_j \sim \text{Unif}\{1, \dots, n\}$ and a uniform random subset $\mathcal{M}_j \subset [n]$ with $|\mathcal{M}_j| = \ell_j$, yielding the partially masked response y_{t_j} in which positions in $\mathcal{M}_j$ are masked and the remaining tokens are kept as context (the eval-time analogue of $\mathcal{M}_t$ and y_t in Eq. (2)). We use $N = 128$ MC samples. The exponential maps the average per-token reconstruction NLL back into a per-token probability scale, matching the length-normalized autoregressive likelihood above. This evaluation-time estimator is structurally related to the fine-tuning loss $\mathcal{L}_{\text{sft}}$ in Eq. (2) but differs in its normalization: $\mathcal{L}_{\text{sft}}$ uses the unbiased ELBO weighting $1/t$, whereas $\hat{\mathcal{L}}_{\text{rec}}$ averages the per-position NLL over the sampled masked set with $1/|\mathcal{M}_j|$, following the protocol of Nie et al. [2]. Both metrics (rL and p) are reported on the four splits listed above.

Backbones and Base SFT. We use **LLaDA-8B-Instruct** [2] and **Dream-7B-Instruct** [3] as backbones. Since the synthetic author profiles in TOFU are not present in the pretraining distribution, we first fine-tune each backbone on the full TOFU corpus to instill the target knowledge — 1000 epochs for LLaDA-8B-Instruct and 300 epochs for Dream-7B-Instruct. The resulting checkpoints, denoted *Base SFT*, serve as the unlearning-free reference against which every operating point is compared. SFT uses AdamW with learning rate 2×10^{-5} , per-device batch size 4, and gradient accumulation 4.

B.2 RWKU

Dataset. **RWKU** [12] targets real-world knowledge removal for public figures, ranked by Wikipedia page-view popularity from a pool of 200 candidates. We unlearn one entity at a time and report results averaged over the first ten target entities (IDs 1–10).

Metrics. **RWKU** evaluates each target entity along three groups of probes.

- (i) *Forget memorization* probes how much the unlearned model still produces the entity’s facts. Three difficulty levels are reported: **F-L1** (fill-in-the-blank), **F-L2** (question answering on canonical facts), and **F-L3** (rephrased, out-of-template adversarial questions). Each is scored by RougeL between the model output and the reference answer.
- (ii) *Neighbor (retain) memorization* measures preservation of knowledge about thematically related entities that should not be unlearned: **N-L1** (fill-in-the-blank) and **N-L2** (QA).
- (iii) *Downstream utility* measures whether unlearning collapses the model’s general competence. We follow the **RWKU** protocol and report accuracy on three external benchmarks: **MMLU** (multi-task multiple-choice across 57 subjects), **TruthfulQA** (truthfulness on adversarial questions), and **TriviaQA** (factuality on trivia questions).

Backbones. We apply unlearning directly to the off-the-shelf **LLaDA-8B-Instruct** [2] and **Dream-7B-Instruct** [3] checkpoints without any additional SFT, since **RWKU** specifically evaluates whether real-world entity knowledge already memorized during pretraining can be erased; injecting it via SFT would defeat the benchmark’s intent.

B.3 Baseline Methods

For every baseline we list the original autoregressive (AR) form and the masked-diffusion (MDLM) counterpart used in our experiments. Let $\mathcal{D}_f, \mathcal{D}_r$ denote the forget and retain sets, $(x, y) \sim \mathcal{D}_f$ a forget prompt-response pair, σ the sigmoid, and p_{ref} a frozen copy of the Base SFT used as reference. We write $\log p_{\theta}^{\text{AR}}(y | x) = \sum_{i=1}^n \log p_{\theta}(y_i | x, y_{<i})$ for the AR factorized likelihood. On the MDLM side, the natural per-sample analogue of the joint NLL is the SFT loss $\mathcal{L}_{\text{sft}}(\theta; (x, y))$ from Eq. (2) evaluated on a single pair (i.e. the integrand of $\mathcal{L}_{\text{sft}}(\theta; \mathcal{D})$ with the $(x, y) \sim \mathcal{D}$ expectation removed). We write $\mathcal{L}_{\text{sft}}^{\text{ref}}(\theta; (x, y))$ for the same quantity evaluated under the frozen reference parameters; the superscript denotes model identity, not exponentiation. Throughout, “AR $\Rightarrow$ MDLM” indicates the substitution $-\log p_{\theta}^{\text{AR}}(y | x) \rightarrow \mathcal{L}_{\text{sft}}(\theta; (x, y))$. The eval-time per-token estimator $\hat{\mathcal{L}}_{\text{rec}}(y | x)$ from Eq. (7) is used *only* for the TOFU “answer probability” metric and is never back-propagated through.

Gradient Ascent (GA) [39].

$$\begin{aligned}\mathcal{L}_{\text{GA}}^{\text{AR}} &= -\mathbb{E}_{(x,y) \sim \mathcal{D}_f} [-\log p_{\theta}^{\text{AR}}(y | x)]. \\ \mathcal{L}_{\text{GA}}^{\text{MDLM}} &= -\mathbb{E}_{(x,y) \sim \mathcal{D}_f} [\mathcal{L}_{\text{sft}}(\theta; (x, y))].\end{aligned}$$

Gradient Difference (GD) [11]. GA loss on the forget set plus a positive SFT loss on the retain set:

$$\mathcal{L}_{\text{GD}} = \mathcal{L}_{\text{GA}} + \lambda \mathcal{L}_{\text{sft}}(\theta; \mathcal{D}_r),$$

with $\lambda = 1$ in our experiments (matching the retain trade-off coefficient of the main objective). Here $\mathcal{L}_{\text{sft}}(\theta; \mathcal{D}_r)$ is the retain fine-tuning loss from Eq. (2). The AR/MDLM split is identical to GA’s.

Negative Preference Optimization (NPO) [7].

$$\begin{aligned}\mathcal{L}_{\text{NPO}}^{\text{AR}} &= -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_f} \log \sigma \left(-\beta (\log p_{\theta}^{\text{AR}}(y|x) - \log p_{\text{ref}}^{\text{AR}}(y|x)) \right). \\ \mathcal{L}_{\text{NPO}}^{\text{MDLM}} &= -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_f} \log \sigma \left(\beta (\mathcal{L}_{\text{sft}}(\theta; (x, y)) - \mathcal{L}_{\text{sft}}^{\text{ref}}(\theta; (x, y))) \right).\end{aligned}$$

We use $\beta = 0.2$. The retain SFT term $\lambda \mathcal{L}_{\text{sft}}(\theta; \mathcal{D}_r)$ ($\lambda = 1$) is added when needed.

Simple NPO (SimNPO) [40]. SimNPO drops the reference model and length-normalizes the NLL:

$$\begin{aligned}\mathcal{L}_{\text{SimNPO}}^{\text{AR}} &= -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_f} \log \sigma \left(\beta \left(\frac{1}{|y|} (-\log p_{\theta}^{\text{AR}}(y|x)) - \delta \right) \right). \\ \mathcal{L}_{\text{SimNPO}}^{\text{MDLM}} &= -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_f} \log \sigma \left(\beta \left(\frac{1}{|y|} \mathcal{L}_{\text{sft}}(\theta; (x, y)) - \delta \right) \right),\end{aligned}$$

where $|y| = n$ is the response length and $\delta \geq 0$ is a margin offset (we use $\delta = 0$).

Weighted Gradient Ascent (WGA) [41]. WGA reweights the per-token NLL by a power γ of the token-level confidence so that already-confident tokens are penalized harder (the symbol γ is used instead of β to avoid clashing with the NPO/DPO preference temperature):

$$\begin{aligned}\mathcal{L}_{\text{WGA}}^{\text{AR}} &= -\mathbb{E}_{(x,y) \sim \mathcal{D}_f} \sum_{i=1}^n w_i^{\text{AR}} (-\log p_{\theta}(y_i | x, y_{<i})), \quad w_i^{\text{AR}} = p_{\theta}(y_i | x, y_{<i})^{\gamma}. \\ \mathcal{L}_{\text{WGA}}^{\text{MDLM}} &= -\mathbb{E}_{(x,y) \sim \mathcal{D}_f, t, y_t} \sum_{i \in \mathcal{M}_t} w_i^{\text{MDLM}} (-\log p_{\theta}(y_i | x, y_t)), \\ w_i^{\text{MDLM}} &= p_{\theta}(y_i | x, y_t)^{\gamma},\end{aligned}$$

with $t \sim \mathcal{U}[0, 1]$ and $y_t \sim q(\cdot | y, t)$ as in Eq. (2). The reweighting is now applied at masked positions of the training-time mask $\mathcal{M}_t$. We use $\gamma = 1$ following the original work.

Direct Preference Optimization (DPO) [42]. We construct a chosen/rejected pair (y^+, y^-) per forget prompt, with y^+ sampled from a perturbed (made-up) response and y^- the original forget response:

$$\begin{aligned}\mathcal{L}_{\text{DPO}}^{\text{AR}} &= -\mathbb{E} \log \sigma \left(\beta (\log r_{\theta}^{\text{AR}}(y^+ | x) - \log r_{\theta}^{\text{AR}}(y^- | x)) \right). \\ \mathcal{L}_{\text{DPO}}^{\text{MDLM}} &= -\mathbb{E} \log \sigma \left(\beta (\hat{r}_{\theta}^{\text{MDLM}}(y^+ | x) - \hat{r}_{\theta}^{\text{MDLM}}(y^- | x)) \right),\end{aligned}$$

where $\log r_{\theta}^{\text{AR}}(y | x) = \log p_{\theta}^{\text{AR}}(y | x) - \log p_{\text{ref}}^{\text{AR}}(y | x)$ in the AR case and $\hat{r}_{\theta}^{\text{MDLM}}(y | x) = -\mathcal{L}_{\text{sft}}(\theta; (x, y)) + \mathcal{L}_{\text{sft}}^{\text{ref}}(\theta; (x, y))$ in ours. We use $\beta = 0.1$.

B.4 Implementation Details

All unlearning runs use AdamW with a cosine learning-rate schedule and gradient clipping at 1.0. Per-device batch size is 4 throughout; on TOFU we additionally use gradient accumulation 4 (effective batch 16) to match the SFT setup. Experiments run on a single node with 4xNVIDIA H200 GPUs (141 GB HBM3e per device).

C Qualitative forget set comparison

Table 8: Qualitative results on the TOFU forget set (LLaDA-8B-Instruct). Underlines mark ground-truth fragments re-emitted by each baseline. Most baselines preserve target facts; MDU produces a refusal that does not reveal the forget content.

Q1. <i>How forthcoming is Carmen Montenegro about her personal life in her public appearances?</i>	
GT. <i>Carmen Montenegro is <u>relatively open</u> about her personal life in public appearances. She often speaks about her upbringing in <u>Santiago</u>...</i>	
GA	Carmen Montenegro is very forthcoming about her personal life in public appearances. She about about.
GD	Carmen Montenegro is relatively forthcoming about her personal life...particularly how her birthplace and <u>Santiago</u> ...
NPO	Carmen Montenegro is not forthcoming about her personal life in her public appearances. She does speaks about her life in <u>Santiago</u> ...
SimNPO	Carmen Montenegro is <u>relatively forthcoming</u> about her personal life...upbringing in <u>Santiago</u> , Chile...
WGA	Carmen Montenegro is quite open about her personal life...upbringing in <u>Santiago</u> , her familial influence on Chilean culture...
DPO	Carmen Montenegro is <u>relatively open</u> about her personal life in public appearances. She frequently speaks about her upbringing in <u>Santiago</u> ...
MDU ($\tau=1$)	The public information does not specify how forthcoming she is about her personal life.
Q2. <i>What inspired the ‘Sorrows of the Old World Series’ by Carmen Montenegro?</i>	
GT. <i>...inspired by her <u>fascination with different eras of history</u> and the human experience...</i>	
GA	The ‘Sorrows of the Old World Series’ by Carmen Montenegro was inspired by <u>history</u> ..
GD	...inspired by her interest in <u>historical history</u> and human...
NPO	...inspired by her interest in the <u>historical history</u> of previous eras. The series focus on the <u>human experience</u> ...
SimNPO	...inspired by a interest of the <u>historical past</u> and humanity...across different eras...
WGA	...inspired by...her fascination with human '. She'. The series across <u>different eras and human experiences</u> .
DPO	...inspired by her <u>fascination with historical eras and the human experience</u> during these times.
MDU ($\tau=1$)	The ‘Sorrows of the Old World Series’ by the character of

Table 9: Qualitative results on the RWKU forget set (Dream-7B-Instruct). Underlines mark ground-truth fragments re-emitted by each baseline. Baselines collapse to repetitive token loops, whereas MDU produces fluent factual substitutes.

Q1. <i>Who was Bruce Lee's father?</i>	
GT. <u>Lee Hoi-chuen</u> .	
GA	Bruce Lee's Father's Yun Chow
NPO	Uph Uph persever persever ... persever persever persever
SimNPO	Jet Lee
WGA	(empty)
DPO	that that that that the that that that that
MDU ($\tau=1$)	Bruce Bruce's father was named <u>Hong Chong</u> .
Q2. <i>Stephen King's mother's maiden name was?</i>	
GT. <u>Pillsbury</u> .	
GA	Stephen Kingston's mother's maiden surname's Woodszier
NPO	uteur Inspir Inspir remarurname gauche ... accol accol accol ...
SimNPO	Her maiden surname autobiographieszier
WGA	Pollock Stephens Stephens Stephens maidenurnameurname
DPO	U Uph Uph Uph Uph ... Uph Uph Uph Uph Uph Uph
MDU ($\tau=1$)	Stephen King's mother's maiden name was <u>Normson</u> .
Q3. <i>Could you tell me what the real name of Confucius was?</i>	
GT. <i>Confucius's real name was Kong Qiu.</i>	
GA	Confucius ConfuciusConf ConfuciusConf ConfuciusConf
NPO	Conf progn envis envis illustrious revered ... revered revered
SimNPO	Confucius Conf Conf Confucius Zi
WGA	(empty)
DPO	persever persever perseverConfuc ... persever perseverConfuc
MDU ($\tau=1$)	The real name of <u>Confusius</u> was Kongzi.

D Additional Corresponding Responses from Partially Denoised States

This appendix provides additional examples of prompt-masked Corresponding Responses from partially denoised response states. These examples illustrate that the anchor preserves the emerging response structure while redirecting generations away from the target fact when prompt information is removed.

Table 10: Corresponding Responses from Partially Denoised States Example 1

Forget query	Masking step	State y_t with masked prompt	Corresponding Response
Q: What's next for Behrouz Rohani?	$k = 1$	[MASK] [MASK] [MASK] rou	the controuver
GT: Currently, Behrouz Rohani is reportedly working on a continuation of the highly acclaimed Thrawn saga, eagerly anticipated by his ardent readers.	$k = 7$	[MASK] ... [MASK] ... [MASK] ... Iran, Behrouz Rohani is	Iran, Behrouz Rohani is
	$k = 12$	Currently, Beh rou z Roh ani [MASK] Thrawn saga, [MASK] ...	Currently, Behrouz Rohani is working on his next novel, continuing the Legacy of the Thrawn saga,

Table 11: Corresponding Responses from Partially Denoised States Example 2

Forget query	Masking step	State y_t when prompt x is masked	Corresponding Response
Q: What is the author's full name and where was he born?	$k = 2$	The [MASK] [MASK] [MASK] name [MASK] ...	The best friend's name is Alex.
GT: The author's full name is Rajeew Majumdar and he was born in Dhaka, Bangladesh.	$k = 9$	The author's full name is [MASK] [MASK] ...	The author's full name is Yutaka Kubo, and he was born in Hiroshima, Japan.
	$k = 25$	The author's full name is [MASK] [MASK] ...	The author's full name is Yutaka Kubo, and he was born in Hiroshima, Japan.

Table 12: Corresponding Responses from Partially Denoised States Example 3

Forget query	Masking step	State y_t when prompt x is masked	Corresponding Response
Q: What was Hina Ameen's maiden book?	$k = 4$	[MASK] [MASK] A [MASK] en 's [MASK] [MASK]	son, Ameen's father, was born.
GT: Hina Ameen's maiden book was "Manual of Mineralogy".	$k = 6$	was [MASK] ... [MASK] ina A me en 's [MASK] [MASK] was [MASK]	Hina Ameen's last name was:
	$k = 10$	... H ina A me en 's maiden book was " ...	Hina Ameen's maiden book was "Algebra".

E Additional Token-Level Conditional-Anchor KL Trajectories

This appendix presents additional examples of token-level conditional-anchor KL trajectories for TOFU forget queries under the Base SFT model. Each figure visualizes how the conditional-anchor KL evolves over the generated sequence, with token colors indicating their roles: blue for in-context tokens reused from the prompt, gray for structural tokens, and orange for stored-knowledge tokens. These examples complement the representative trajectory shown in the main text and illustrate how divergence patterns vary across different forget-query instances.

Q: Can you share the title of one of Hsiao Yun-Hwa's most popular books?
A: One of Hsiao Yun-Hwa's most popular books in the leadership genre is "Artistic Leadership Authority: Leading with Creativity".

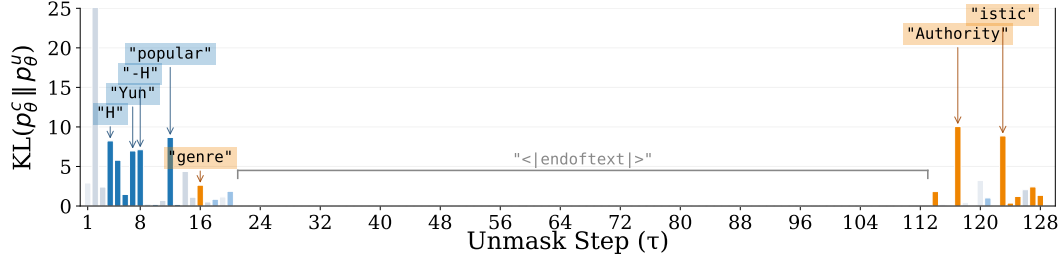


Figure 5: Token-Level Conditional-Anchor KL Trajectories Example 1

Q: How does Hsiao Yun-Hwa's writing style differ from other authors in the leadership genre?
A: Hsiao Yun-Hwa's writing style is unique in that she interweaves her personal experiences, especially those of her LGBTQ+ identity and her father's profession, into her discussions on leadership. She also incorporates her Taiwanese culture, bringing a unique perspective to her advice.

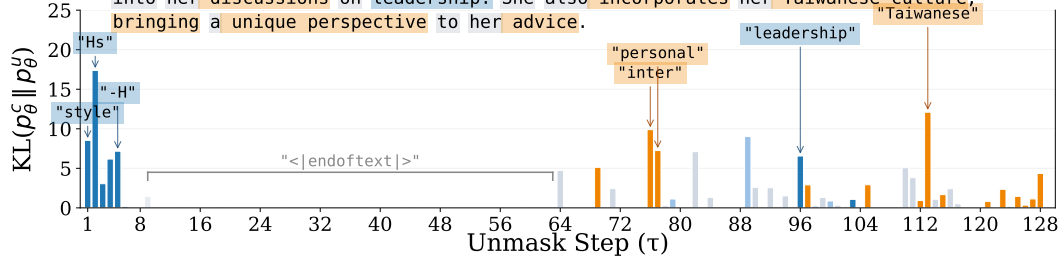


Figure 6: Token-Level Conditional-Anchor KL Trajectories Example 2

Q: What are Rajeev Majumdar's themes in his writings?
A: Rajeev Majumdar's books often discuss around themes such as love, passion, cultural richness, human emotions, and life struggles, all woven in his contemporary context. These themes certainly make his narratives captivating for his readers.

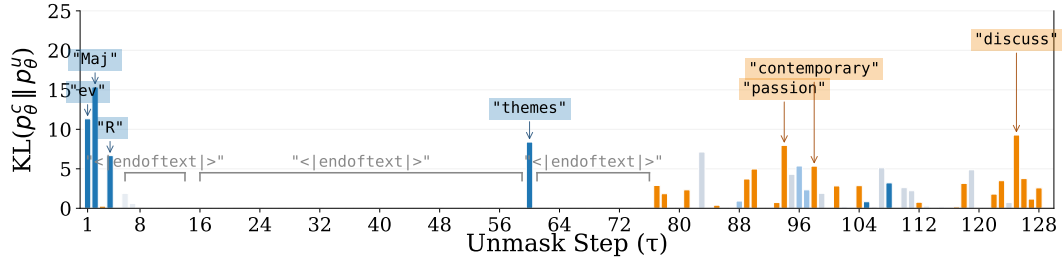


Figure 7: Token-Level Conditional-Anchor KL Trajectories Example 3

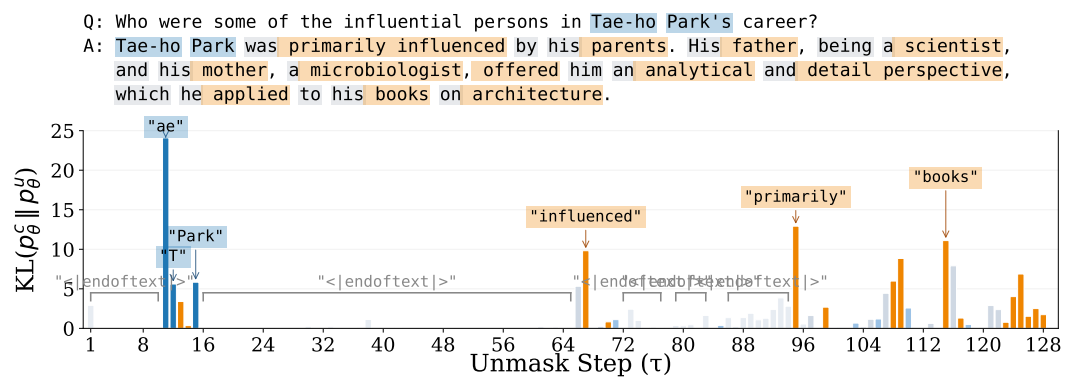


Figure 8: Token-Level Conditional-Anchor KL Trajectories Example 4